

PROGRAM 3: Code a dockerized python flask or [Node.js](#) application

Directory Structure:

>>[app.py](#)

>>Dockerfile

>>requirements.txt

Step 1: Create a Dockerfile as following

```
GNU nano 7.2 Dockerfile
FROM python:latest
#Initialize Working Directory
WORKDIR /app
COPY requirements.txt .
#Install Dependencies
RUN pip install --no-cache-dir -r requirements.txt
#Copy the Source file
COPY . .
#Expose the Port 5000
EXPOSE 5000
#Run the app file
CMD ["python", "app.py"]
```

Step 2: Create [app.py](#)

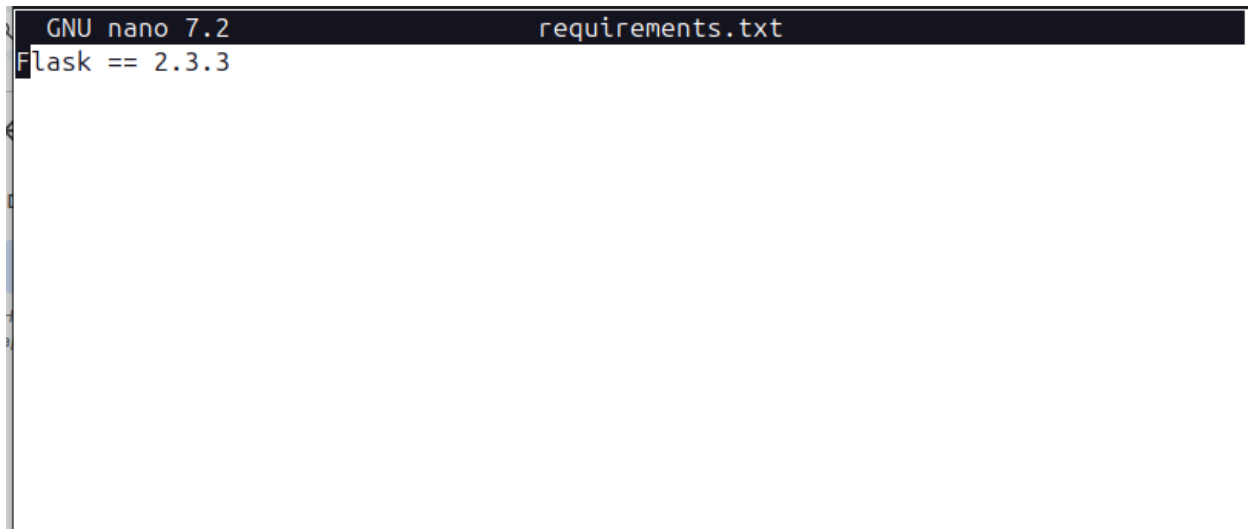
```
GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Dockerized Flask Application"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Step 3: Create requirements.txt



Step 4: Build the docker image

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog3$ sudo docker build -t prog3 .
[sudo] password for 1RV24MC025_CHANDAN_K:
[+] Building 84.5s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 306B
=> [internal] load metadata for docker.io/library/python:latest
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fcd62698eb9a9c32257516538a2746c19a
=> resolve docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fcd62698eb9a9c32257516538a2746c19a
=> sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca 25.61MB / 25.61MB
=> sha256:6d58c1a9444bc2664f0fa20c43a592fcd62698eb9a9c32257516538a2746c19a 10.95kB / 10.95kB
=> sha256:3b700cf7651a764f8c33f574fe4f71be54f77cf626c3b2572e26493ca465e3b1 2.32kB / 2.32kB
=> sha256:1edf53fe028130609b08c05371a2ad9c5ac12e7474dc0cd6c4902c7eb33a9c9 6.49kB / 6.49kB
=> sha256:378c64c4458002be86f2d86c5768ae9ec0c971f00a43e3d3cda71ffe76a 29.40MB / 29.40MB
=> sha256:281b08c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e 49.29MB / 49.29MB
=> sha256:02e37ab5c33a09f86edcf3602251a61a6208ddaa131e1a1d5757e5f94a54d645 235.98MB / 235.98MB
=> extracting sha256:281b08c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e 1.2s
=> sha256:b1741a62ceed4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d 6.08MB / 6.08MB
=> sha256:add107facb26b82200d59d5550cbbab6af9cf38c971f00a43e3d3cda71ffe76a 29.40MB / 29.40MB
=> extracting sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca 0.5s
=> extracting sha256:378c64c4458002be86f2d86c5768ae9ec0c971f00a43e3d3cda71ffe76a 1.8s
=> sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbd48d468d 249B / 249B
=> extracting sha256:02e37ab5c33a09f86edcf3602251a61a6208ddaa131e1a1d5757e5f94a54d645 4.5s
=> extracting sha256:b1741a62ceed4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d 0.2s
=> extracting sha256:add107facb26b82200d59d5550cbbab6af9cf38c971f00a43e3d3cda71ffe76a 0.5s
=> extracting sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbd48d468d 0.0s
=> [internal] load build context
=> => transferring context: 93B
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> writing image sha256:89b5089c3a208336b34d3eb5941c3a00b0721bf3dc238ae4ca3556818061e2ec
=> naming to docker.io/library/prog3
```

Step 5: Run the docker image

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog3$ sudo docker run -p 5000:5000 prog3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.3:5000
Press CTRL+C to quit
172.17.0.1 - - [05/Jan/2026 10:19:42] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [05/Jan/2026 10:19:42] "GET /favicon.ico HTTP/1.1" 404 -
```

Step 6: Go to <http://127.0.0.1:5000/> or <http://172.18.0.2:5000/> in the browser

